

Hanbin Zheng

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Education

Zhejiang University - University of Illinois Urbana-Champaign Institute (Dual degree program):

University of Illinois Urbana-Champaign, Illinois, US Sep 2024 – Jun 2028
BS in Computer Engineering, GPA: 3.985/4.0

Zhejiang University, Zhejiang, China Sep 2024 – Jun 2028
BS in Electronic and Computer Engineering, GPA: 3.899/4.0

Awards and Honors

First Prize in ZJU Mathematical Modeling Competition Jun 2025
Zhejiang Provincial Scholarship Oct 2025
Zhejiang University Second Scholarship Oct 2025
Third Prize in National RoboMaster University League (Shanghai, 3v3) Apr 2026

Research Experience

Undergraduate Researcher, ZJUI Jun 2025 – Dec 2025

Advisor: **Prof. Xiangming Meng**

- **Flow Models Exploration** 
Conducted an in-depth study of Flow Matching generative modeling, implementing conditional image generation on MNIST and CIFAR-10 from scratch.
- **MeanFlow for Image Inverse Problem** 
Investigated one-step generative modeling with the recently proposed MeanFlow models, altered its initial distribution from Gaussian distributions to noisy images, achieving efficient one-step denoising.
- **GMFlow for Inverse Problem** 
Investigated Variational Flow Matching and Gaussian Mixture Flow Matching for image inverse problem; implemented 2D toy examples to validate our new algorithm.

Selected Projects

RoboMaster Hero Robot Electrical Control System  Nov 2025 – Apr 2026

- **Embedded Control:** Developed real-time control system on STM32H723 for a steering wheel hero robot, managing high-frequency control loops and peripheral hardware integration.
- **Orientation Estimation:** Independently derived **quaternion kinematics** and implemented the **Mahony Filter** from scratch for IMU sensor fusion, achieving stable orientation tracking.

Deep Reinforcement Learning  Jul 2026 – Present

- **Curriculum & Implementation:** Self-studied UC Berkeley CS285 (Deep RL), implementing core algorithms from scratch including Behavioral Cloning (Flow Matching Policy), Policy Gradients, Actor-Critic methods, and Deep Q-Learning.
- **Experimental Evaluation:** Benchmarked agent performance and trained policies across diverse environments such as PushT, InvertedPendulum, and LunarLander.

Skills

Programming Languages: Python (PyTorch), C/C++, LaTeX

Development Tools: Git, Linux, Makefile, CMake, Vim/NeoVim, Emacs

Deep Learning: Unet, Transformer, ViT, DiT

Generative Modeling: Flow Matching, MeanFlow, Diffusion, GMFlow

Robotics & Embedded: STM32, Basic ROS2/ROS2 Control, Quaternion & Euler Angles, Soldering

Reinforcement Learning: Behavioral Cloning, Model-Free RL (Policy Gradient, Actor-Critic, DQN)